

$$\max[(10\text{-year CMS} - 2\text{-year CMS}) - \text{Strike}, 0]. \quad (9.3)$$

The buyer of a CMS curve cap is taking the view that the 2s10s curve is too flat (on a forward basis) and will steepen over time. Conversely, CMS curve floors are used to take the view that the curve is too steep (on a forward basis) and will flatten over time. A linear curve floor on 2s10s has a payoff that depends on

$$\max[\text{Strike} - (10\text{-year CMS} - 2\text{-year CMS}), 0]. \quad (9.4)$$

Example

Consider the curve option trade whose terms are shown in Table 9.2. In this trade, the client pays an upfront premium of 80 bps of the notional, or \$800,000, to the dealer. Starting in three months and every three months thereafter over the life of the trade, both 30-year CMS and 2-year CMS are reset. For any settings that are in-the-money (meaning, the difference between 30-year CMS and 2-year CMS is above the strike), the associated payment is made three months later. Final payments (if any) occur five years from the start of the trade. $\square$

When trades like these were first being actively marketed to investors, the terms of trades such as the one shown in Table 9.2 were often accompanied by graphs like the one in Figure 9.6. This figure shows a long-term history of the 2s30s swap curve with the strike and breakeven on the trade superimposed. The 2s30s curve needs to be in excess of 47 bps on a reset date in order for the option to set in-the-money. Incorporating the upfront premium into the trade, an average setting of 65 bps on each reset date is the level necessary to break even on the trade.¹⁹ As shown in this figure, there were periods of time in which 2s30s was far steeper than that, making this an attractive trade to many.

Note that with this particular curve option, the buyer is actually purchasing a series of quarterly options, much like the buyer of a cap or floor as we discussed in Chapter 4. Sometimes linear curve option trades were structured such that the buyer would receive a payout depending only on a single reset. As we will see in Section 9.3.1, other curve option trades are structured

¹⁹If the 2s30s curve were to set at the strike of 47 bps every period, then there would never be any payoff from the option, and the net loss on the trade would be the 80 bps upfront premium paid for the option. If the curve 2s30s sets at 65 bps every quarter, then the buyer of the option would be receiving an annuity payout of 65 bps – 47 bps = 18 bps, paid quarterly for five years. The present value of this annuity is 80 bps, sufficient to offset the upfront premium for the option.

Buyer:	Client
Seller:	Dealer
Maturity:	5 years
Resets:	Quarterly
Payments:	Quarterly
Strike:	47 bps
Premium:	80 bps of the Notional
Notional:	\$100mm
Payments:	$\max[(30\text{-year CMS} - 2\text{-year CMS}) - \text{Strike}, 0]$ $\times \text{Notional} \times 90/360$

Table 9.2: Terms of Curve Cap on 2s30s

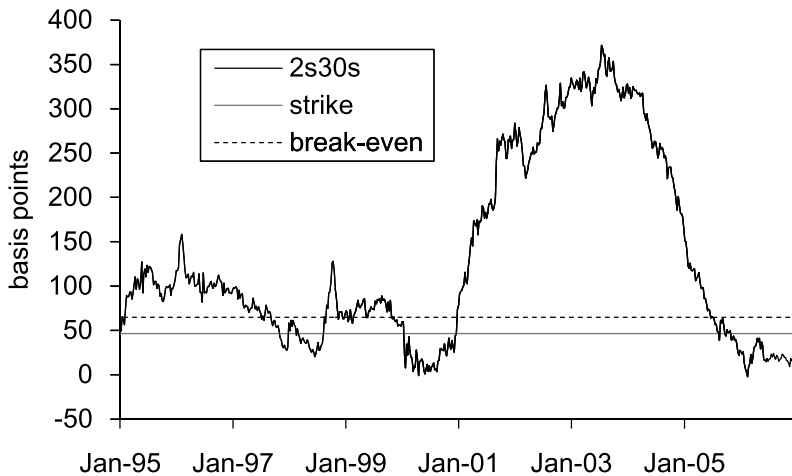


Figure 9.6: History of 2s30s with Trade Analysis

with binary payouts, in which case the buyer of the option would receive either a fixed payment or zero, depending on what happens to the underlying curve.

Trades like these became very popular with investors in 2005 and 2006. In 2004, products like these were nearly nonexistent. According to Sawyer (2005), by the end of 2005 nearly \$30 billion of these types of trades had been printed, and some market participants have suggested that by 2007 hundreds of billions were in existence.

9.3.1 Why Did Curve Options Come About?

In Chapter 7 we took a look at a very popular structured note called a *non-inversion note*. As a reminder, let's take a look at a non-inversion note whose terms are found in Table 9.3.²⁰

What's going on here? The investor who is buying this note is receiving an (above-market) fixed coupon of 7% for the first three years. Thereafter, the investor will accrue this high fixed coupon each day that the 2s10s swap curve does not invert. If, however, the 2s10s curve inverts on any given day, the investor accrues a 0% coupon for that day instead of a 7% coupon. To put this differently, for the last seven years of this bond, the investor is receiving a 7% coupon and is short some options. The options the investor is short are binary in nature: they pay out a 7% coupon any day the 2s10s curve inverts, and zero otherwise. As we discussed in Chapter 7, the dealer that does the underlying swap transaction in conjunction with this note is getting long these options when the note is created.

The nature of the options that the dealer is getting long is depicted in Figure 9.7. In this figure, S denotes 10-year CMS – 2-year CMS. On an expiry date, when $S \geq 0$, the option has zero payout. However, when $S < 0$, the option pays out a fixed amount equal to the 7% fixed coupon of the bond. Given the terms of the note in Table 9.3, the dealer is actually purchasing a series of binary options of this form that expire daily, the first of which expires in three years and the last of which expires ten years after trade inception.

An Attractive Curve for Buying Non-Inversion Notes

The years 2005 and 2006 were marked by two very notable characteristics of the swap curve. As shown in Figure 9.8, the curve (as measured here by 2s10s) was getting very flat, and the curve was becoming less volatile. The flatness of the curve was thought to be driven primarily by the activities of pension funds, which were known to be actively buying long-dated assets and receiving in long-dated swaps. As the curve flattened dramatically (and the curve on a forward basis also flattened), non-inversion notes became more and more attractive. With the flattening of both spot and forward curves (as shown in Figure 9.8), pricing models placed a greater chance on the curve inverting, an event whose occurrence would mean that the buyer of a non-inversion note would get a zero coupon. As compensation for bearing this risk, the fixed coupons that could be offered in newly issued

²⁰Many non-inversion notes that have been printed are callable, meaning the issuer has the right to call the bond from investors after some period of time. The existence of this call feature enhances the coupon that can be paid to the bondholder. For simplicity, the fictitious example that we consider here is assumed to be non-callable.

Principal:	\$100mm
Issuer:	U.S. Corporate
Maturity:	10 years
Coupons:	Years 1–3: 7% paid quarterly 30/360 Years 4–10: $7\% \times N/M$ paid quarterly 30/360 where N = number of days in quarter that Curve ≥ 0 M = number of days in quarter
Curve:	10-year CMS – 2-year CMS

Table 9.3: Terms of a Non-Inversion Note

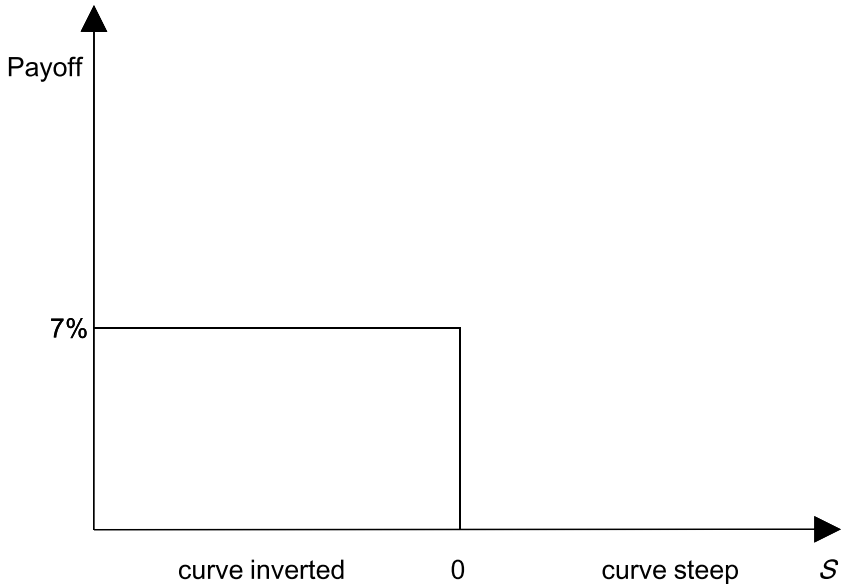


Figure 9.7: Payoff of a Binary Curve Option at Expiry

non-inversion notes were higher than what could be obtained in a steeper curve environment, making these bonds even more attractive to investors who believed that the 2s10s curve in swaps would never invert. Despite the tremendous flattening observed, the swap curve had historically been very resistant to any inversion. This made some investors comfortable taking the risks inherent in non-inversion notes.²¹

²¹The 2s10s swap curve did invert in 2006, first in February and again in November, but the inversion in each case was short-lived.

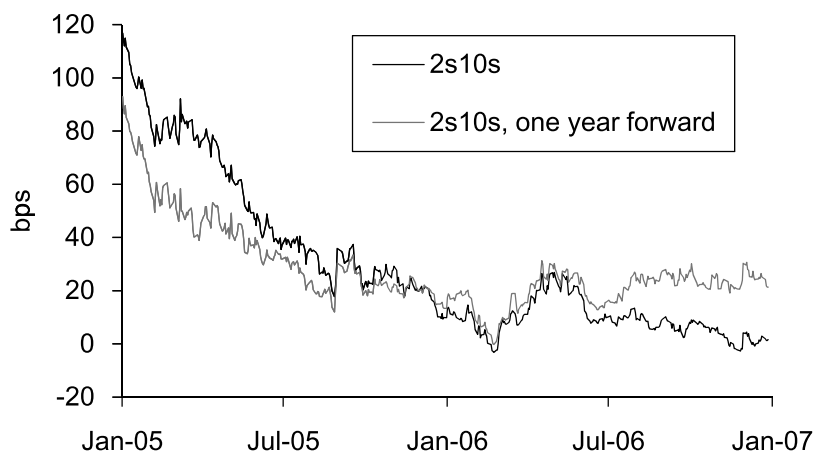


Figure 9.8: 2s10s Swap Curve, Both Spot and One Year Forward, in 2005–2006

Historically, a flat curve has also been associated with low levels of realized curve volatility. This was certainly true in 2006. Typically when the curve is flat, it is a period of time in which many market participants believe the Fed to be on hold, as was the case during this period of time. In addition, dealers were getting long curve options from all of the non-inversion notes that were being printed and, in many cases, had to maintain their long position in these options as a liquid secondary market for such options was then unavailable. Dealers who owned these options delta hedged them, which meant that they had to put on outright flatteners or steepeners as the curve moved. As these binary options were out-of-the-money (forward), the delta hedging activities by dealers who were long these options meant that they would put on steepeners as the curve flattened and put on flatteners as the curve steepened. This meant that, on the margin, the delta hedging of dealers actually served to reduce volatility in the curve during this time. This factor is thought by many market participants to have greatly contributed to the low level of realized volatility in the curve during this time.²²

²²Most of these binary curve options that dealers implicitly purchased through the issuance of non-inversion notes were “low-strike” options. For example, in the non-inversion note in Table 9.3, the embedded binary options are struck at zero. As suggested by Figure 9.8, these binary options were thus struck out-of-the-money (forward).

Consider now an option trader who is long a put option on a stock and is delta hedging the option. As the stock price goes down, the option trader will buy the stock to delta hedge, which puts (marginal) upward pressure on the stock price. As the stock price goes up, the option trader will sell the stock to delta hedge, which puts (marginal) downward

9.3.2 Implied Correlation

Let's consider again the nature of the options that the dealer is buying when the non-inversion note in Table 9.3 is issued. Define $S = 10\text{-year CMS} - 2\text{-year CMS}$. Consider an option that has a daily look each day for a period of seven years, starting in three years. The daily accrual from the option is as follows:

$$\text{Payoff} = \begin{cases} 7\% & \text{if } S < 0 \\ 0 & \text{otherwise.} \end{cases} \quad (9.5)$$

Based on these daily accruals, payments from the option are made at the end of each quarter over this seven-year period. The payment made at the end of each quarter is a function of the quantity

$$\text{Notional} \times 7\% \times N/M,$$

where N is the number of days in the quarter that $S < 0$, and M is the number of days in the quarter.

This curve option is binary in nature (as compared to the curve options with linear payouts that we looked at in Equations 9.3 and 9.4). Furthermore, it is referred to as a *curve floor* because it pays out when the curve, as measured by S , finishes below some strike (in this case, 0).

What's the volatility of the underlying, S ? Its variance is given by

$$\sigma_S^2 = \sigma_{10Y}^2 + \sigma_{2Y}^2 - 2\rho_{2Y,10Y}\sigma_{10Y}\sigma_{2Y}, \quad (9.6)$$

where $\rho_{2Y,10Y}$ is the *implied correlation* between the 2-year swap rate and the 10-year swap rate. Thus, the implied volatility of the curve, S , depends

pressure on the stock price. Thus, the delta hedger of this put option actually serves to stabilize the price of the underlying on the margin. If the option position is large enough, the delta hedging of the option position could actually exert significant pressure that would cause the underlying stock price to remain remarkably nonvolatile. See Appendix A for a review of these dynamics.

This is precisely what happened as the supply of curve options mounted at dealers, and they delta hedged these positions. In the case of curve options, to delta hedge means to either put on a steeper or flattener as the curve changes shape. As dealers were long out-of-the money binary curve options, as the curve steepened, they put on flatteners. And as the curve flattened, they put on steepeners. Thus, the delta hedging of curve options by dealers tended to lower the realized vol of the curve. The large position of curve options dealers owned and the related delta hedging of these options were thought to be a major force behind the limited curve movements observed in 2006.

The nature of the delta hedging activities of dealers who are long binary curve options depends crucially on whether or not the options are in-the-money or out-of-the-money. As indicated, when these options are out-of-the-money, the delta hedging activities of dealers serves to limit curve movements. As we shall discuss, the delta hedging activities by dealers who are long these options dramatically changes and serves to exacerbate curve moves when these options are in-the-money.

not only on the implied volatility of 10-year CMS and the implied volatility of 2-year CMS but also on the correlation between them. As suggested by Equation 9.6, implied volatility of the curve and implied correlation of the curve have an inverse relationship; as the implied volatility of the curve goes up, this must mean that the implied correlation of the curve must go down (all else equal), and vice versa.

Thus, the proliferation of this type of option introduced a new type of risk; not only were dealers and investors exposed to the volatility of the 2-year swap rate and the 10-year swap rate, but they were also exposed to the correlation between them. We refer to this type of exposure as *correlation risk*. There was not a well-formed market for correlation risk in the rates markets at this time, and often dealers had to live with this correlation risk in their books, as there were limited avenues to get out of it.

9.3.3 Implied Volatility Versus Realized Volatility

We have thrown around two concepts of volatility, implied and realized, and should be careful to distinguish between the two. Implied volatility refers to the volatility computed from option prices. In the case of curve options, as the supply of curve options increased as more and more non-inversion notes were printed, the price dealers were willing to pay for these options went down. And as the price of curve options went down, this meant that implied volatilities went down as well.²³ Recall again Equation 9.6, which shows the relationship between the variance of the difference of two random variables and their correlation. From this equation, we see that as implied volatility goes down, this must mean that implied correlation must go up.²⁴

The implied volatility of curve options depends on the expiry of the option. A typical relationship often found in 2005–2006 between the implied vol of a curve option and its expiry is shown in Figure 9.9. As shown in this figure, short-dated curve options usually traded at implied volatilities that were higher than those found with longer-dated curve options.²⁵ There are

²³As discussed in Appendix A, the vega of binary options can be either positive or negative, depending on the moneyness of the option. In the case of binary curve options written in newly issued non-inversion notes, the options were out-of-the-money, and as such they had positive vega and a decrease in the price of the option was associated with a decrease in implied vol.

²⁴This is not as mysterious as it may seem. As implied volatility of the curve goes down, this means that the curve, say, 2s10s, is not expected to move a lot. If the curve is not moving around a lot, this means that the 2-year swap rate and the 10-year swap rate must be moving together. That is to say, there is high correlation between the two rates. Low implied volatility of the curve implies high implied correlation, and vice versa.

²⁵The pattern observed here, with short-dated curve options trading at a higher implied volatility than longer-dated options, tended to be much more pronounced than what was typically observed in most other rate option markets.

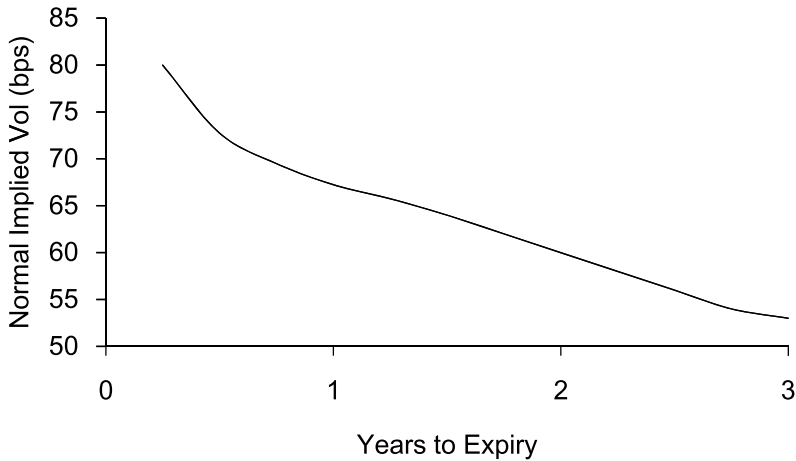


Figure 9.9: Implied Volatility of Curve Options on 2s10s Typically Observed in 2005–2006

two reasons often cited for this pattern. First and foremost, there tended to be relatively little supply of short-dated curve options from non-inversion notes. Most of these notes had fixed coupons for at least a year, and thus the options embedded in these notes tended to be longer dated. For example, the note in Table 9.3 has fixed coupons for three years, and the curve options have expiries of three years and longer. The supply of long-dated curve options depressed the vol of these options as we have discussed, certainly relative to scarcer short-dated curve options. Second, some have suggested that the downward-sloping shape of the vol curve for curve options is consistent with the notion of mean reversion of interest rates.

Realized volatility (sometimes called *actual volatility*) refers to the empirically observed volatility of a traded variable. As mentioned, due to dealer delta hedging of curve options in 2005–2006, realized vol of the curve was decreasing over time. This can be seen by again looking at Figure 9.8; not only was the curve flattening significantly over this time, but the realized volatility of the curve was decreasing as well. For example, in 2006 the 2s10s curve stayed between 0 bps and 20 bps most of the year, whereas in the prior year it had traded in a range in excess of 100 bps.

In Section 4.1.5 we said that the name of the game for dealers that make markets in options is determining what vol assumptions to use when pricing these options. That is only half of the game. The other half of the game has to do with implied volatility versus realized volatility. When dealers buy or sell options, they often delta hedge them. A dealer who sells an option

takes in premium today and will delta hedge the option based on the realized moves in the underlying over the life of the option. Assuming the option has positive gamma, delta hedging will be costly. The premium can be thought of as compensation for any costs of delta hedging over the life of the option. A dealer who sells an option and delta hedges is therefore hoping for muted moves in the underlying.

On the other hand, a dealer who buys an option pays a premium and will also delta hedge based on realized moves in the underlying over the life of the option. In this case delta hedging the option based on realized moves will provide profits to the dealer, again assuming the option has positive gamma. A dealer who buys an option and delta hedges is therefore hoping for large realized moves in the underlying.

Thus as dealers buy and sell options, determining the implied volatilities at which they are willing to trade, they are also implicitly expressing a view on the level of implied volatility versus realized volatility. Those who think that there will be relatively low realized volatility in the underlying will want to sell options to take in premium and hope to experience low delta hedging costs over the life of the option. Those who think that there will be high realized volatility in the underlying will prefer to buy options and delta hedge, expressing the view that their profits from delta hedging will exceed the cost of the option.

9.3.4 Supply and Demand of Curve Options

As more and more non-inversion notes were printed in 2005–2006, dealers were getting longer and longer binary curve floors. There were a number of underlying curve points that were popular: 10s30s was a very popular choice, representing by some estimates approximately 40% of all non-inversion issuance during this period, with 2s10s and 2s30s representing a bulk of other types (see Patel [2006]). In addition, notes were written with a variety of different strikes. Although a strike of zero was a popular choice, other strikes existed as well. For example, some investors looking to buy non-inversion notes wanted a bit more protection than afforded by having a strike of 0 bps and were interested in notes with lower strikes such as -20 bps.

In general, though, the theme was that through the surge of issuance of non-inversion notes, dealers were getting long low-strike binary curve floors. The best way to hedge this position would just be to sell it, but oftentimes it was hard to find investors willing to buy low-strike binary curve floors. Dealers had a bit more success finding investors willing to buy low-strike linear curve floors but often had to move the strike up a bit in order to entice investors to buy such a thing. For example, if a dealer had bought a binary

curve floor on 2s10s struck at -20 bps through the issuance of a non-inversion note, it might be able to find an investor willing to buy a linear floor on 2s10s struck at $+20$ bps. Although this left the dealer with some strike mismatch as well as a mismatch between binary and linear options, at least it enabled the dealer to get out of some of the correlation risk it had obtained.

As investors became more familiar with curve options, dealers tried to sell a variety of low-strike curve floors. But the real demand for the product was in high-strike curve caps. There were many investors in 2006 and 2007 who were expecting the curve to steepen dramatically. As 2s10s hovered around 0 bps in 2006, many investors were interested in buying, for example, curve caps on 2s10s struck at $+50$ bps or $+100$ bps that expire in a year or more. These options were very far out-of-the-money (both on a spot basis and forward basis) and were therefore cheap. Some investors, most notably hedge funds, would view those options as cheap “lottery tickets” (and indeed some of these investments did pay off in the years to follow when the curve steepened dramatically). Demand for these high-strike curve caps was strong.

In summary, in 2005 and 2006 dealers found themselves in the position of being long a lot of low-strike curve floors and being short a lot of high-strike curve caps. The curve floors they were long were often near-the-money, while the curve caps they were short were out-of-the-money. Sitting on such strike mismatch was not a great position for dealers to be in. But in most cases, this was the best that dealers could do in order to get out of some of the correlation risk they were exposed to by issuing non-inversion notes.

9.3.5 The Pricing of Curve Options

The pricing of curve options is complicated due to the correlation of the two underlying points of the curve.²⁶ As more dealers started pricing curve options in 2005, it was common to see a wide variation in prices shown by different dealers. This called into question the types of methodologies that different dealers used to price these options.

Let’s consider again a linear curve option on 2s10s and define $S = 10\text{-year CMS} - 2\text{-year CMS}$. As discussed in Sawyer (2005), dealers used essentially one of the following three methodologies (presented in order of increasing complexity and sophistication) to price curve options:

- Methodology 1: Regard S as a primitive random variable in and of itself (i.e., do not regard S as the difference between two random variables

²⁶Actually, it’s more complicated than that. Recall that in Section 6.5 we discussed that a convexity adjustment to the forward swap rate is required if a counterparty is paying a CMS rate. Similarly, the forwards used to price options on CMS rates should also be convexity adjusted. We will ignore this fact throughout this chapter.